



AI-assisted drawing therapy for children with ADHD: A randomised controlled trial[☆]

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ABSTRACT

Objective: This randomized controlled trial evaluated the efficacy of AI-assisted drawing therapy for children with ADHD, comparing its impact on core symptoms and functional impairments against traditional drawing interventions while assessing engagement and the therapeutic outcomes.

Methods: Forty-one children (ages 7–10) with ADHD were randomized into AI-assisted (n = 19) or traditional drawing therapy (n = 22) groups (Trial registered in ChiCTR: ChiCTR2400089640). The AI intervention employed a process that captured children's drawings, converted them into keywords, generated AI responses based on those keywords, and provided feedback to the children, while the control group received therapist-guided verbal feedback. SNAP-IV-26 (Swanson, Nolan and Pelham-IV rating scales) and WFIRS-P (Weiss Functional Impairment Rating Scales-Parent) scales assessed at baseline (T1) and post-intervention (T2) at a 24-week interval, with data analyzed using independent t-tests (p < 0.05).

Results: Both AI-assisted and traditional drawing therapies significantly reduced core ADHD symptoms. In the before-and-after comparison within the group, the AI group showed targeted improvements in attention (p = 0.039) and oppositional defiant behaviour (p = 0.025), while traditional therapy produced broader symptom reductions across all domains (p < 0.05). Between-group comparisons showed that the AI-assisted group achieved greater improvement in inattention ($\Delta = -1.09$, p = 0.035), whereas the traditional drawing group improved more on oppositional defiant behavior ($\Delta = +1.706$, p = 0.010) and the SNAP-IV total score ($\Delta = +4.777$, p = 0.041). On the WFIRS-P, the AI group showed advantages for the Family domain (p = 0.041) and the total score (p = 0.047), while Self-Concept favored the traditional drawing group (p = 0.036).

Conclusion: AI-assisted, drawing-based art therapy produced greater improvement in the pre-specified primary outcome (SNAP-IV inattention) and select WFIRS-P domains (e.g., Family, total score), whereas conventional drawing therapy yielded broader reductions in oppositional defiant symptoms and the overall SNAP-IV severity. Effects were modest and domain-specific; larger, multi-center trials are needed to confirm efficacy and identify which children benefit most.

Abbreviations: AI, Artificial Intelligence; ADHD, Attention Deficit Hyperactivity Disorder; CBT, Cognitive Behavioural Therapy; DT, Drawing Therapy; AIGC, Artificial Intelligence Generated Content; SNAP-IV-26, Swanson, Nolan and Pelham-IV rating scales 26; WFIRS-P, Weiss Functional Impairment Rating Scale – Parent.

[☆] The trial is registered in the Chinese Clinical Trial Registry (ChiCTR2400089640).

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1. Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental disorder in children, characterized by inattention, hyperactivity, and impulsivity.¹ According to the World Health Organization,² the global prevalence of ADHD is approximately 3.1 % among children aged 10–14 and 2.4 % among adolescents aged 15–19, with boys diagnosed three times more often than girls.³ These symptoms often lead to persistent functional impairments in academic, social, and family domains.⁴ While current first-line treatments include pharmacotherapy and Cognitive Behavioral Therapy (CBT),⁵ both approaches face limitations such as side effects, high cost, limited long-term efficacy, and poor adherence.^{6,7} In response to these challenges, non-pharmacological interventions such as drawing therapy (DT) have gained traction. DT, involving stages of creation, guided sharing, and evaluation, allows children to express internal states non-verbally, facilitating psychological insight and emotional regulation.^{8–10}

Studies have shown its effectiveness in treating psychiatric conditions like Anxiety,¹¹ autism spectrum disorder (ASD), and Alzheimer's disease.^{12,13} While some evidence supports its potential for reducing anxiety in children with ASD,¹⁴ DT remains underexplored in the context of ADHD. Moreover, its reliance on sustained attention and engagement presents particular challenges for children with ADHD, who may struggle with persistence and motivation. Despite its benefits, drawing therapy demands sustained attention and patience—challenges for children with ADHD. It is worth considering how to better guide the practice of drawing therapy in children with ADHD.

Recent advances in artificial intelligence, particularly in Artificial Intelligence Generated Content (AIGC), offer promising enhancements to traditional art therapy. AIGC technologies can generate creative content—such as images and music—through models that interpret textual prompts, thereby producing high-quality, personalized outputs with minimal side effects.^{15,16} Diffusion models^{17,18} have been successfully applied in generating high-resolution art, potentially addressing the attention and motivation barriers in ADHD by making creative activities more engaging. These developments have sparked interest in using AI-generated art in mental health interventions, particularly for children with neurodevelopmental disorders.^{19,15–18,20,21} This technology shows great potential in assisting art therapy, with minimal side effects, and can provide personalized treatment options for cognitive functions such as attention.^{20,22}

Currently, by integrating AI-generated images into drawing therapy assessments and applying traditional statistical methods to evaluate their effectiveness, researchers are working to improve the accuracy of intelligent image evaluation.^{22,23} Building upon this progress, AI-assisted drawing therapy specifically attempts to overcome the challenges inherent in traditional drawing therapy for pediatric patients.²⁴ For instance, it addresses issues like distraction and lack of training persistence through immediate visual feedback, thereby enhancing children's intrinsic motivation and self-efficacy.²⁵ Moreover, multiple randomized controlled trials have confirmed that combining behavioral therapy with psychological interventions significantly enhances executive function in children with ADHD and reduces behavioral deficits in adolescents.^{26–28} Artificial intelligence increases the appeal of drawing therapy by boosting patient engagement and willingness to participate. This offers practical guidance for promoting ADHD patients' involvement, though professional team support remains essential for achieving optimal outcomes.^{29–31}

Despite the growing interest in AIGC-assisted art therapy, few studies have systematically evaluated its impact on ADHD-related symptoms and functional impairments. The integration of AI-generated images into drawing therapy may improve engagement and long-term adherence, yet its efficacy remains empirically under-investigated. To bridge this gap, the present randomized controlled trial aims to:

(1) Examine the effect of AI-assisted drawing on core ADHD symptoms in children;

(2) Assess its influence on functional impairments across daily contexts;

(3) Explore how AI-generated images can enhance therapeutic engagement and personalization in future ADHD interventions.

2. Materials and methods

2.1. Ethics approval

This study adhered to the Declaration of Helsinki and was approved by the [University Ethics Committee, approval MUST-20230925001] and Zhuhai Maternal and Child Health Hospital (ZHFY-KXX-2023038; 2023 Ethics No. 175; 2023–11–08). The trial was registered in the Chinese Clinical Trial Registry (ChiCTR2400089640). In addition, the children's drawings were de-identified by name and date and stored by number, and the original data were encrypted and stored in a dedicated server in the hospital, accessible only to the research team.

2.2. Trial design

This randomized controlled trial, which officially commenced in 2023 and was scheduled to last for six months, involved guardians of participants completing the SNAP-IV-26 (Swanson, Nolan and Pelham-IV rating scales) and WFIRS-P (Weiss Functional Impairment Rating Scales-Parent) questionnaires to obtain baseline values (T1) before the experiment, with the same questionnaires administered upon completion to acquire post-test data (T2) at a 24-week interval between T1 and T2. We started recruitment in early September and completed the recruitment of subjects and the completion of the required informed consent forms within two weeks. During this trial, none of the children included in the trial required additional therapeutic medications. The intervention groups included AI-assisted intervention and traditional drawing intervention, conducted simultaneously weekly. Each weekly session (20–30 min per session, 24 lessons in total) followed a standardized protocol, consistent with behavioral therapy guidelines. In line with prior work, the first 15 min of drawing therapy are considered crucial for effectiveness.³² Additionally, participating physicians served as psychotherapists during sessions (clinical observation only). All outcome scales were collected by a separate, blinded physician assessor who had no role in delivering the interventions. Art therapists guided the creative process and observed behaviors, while technical staff handled the AI generation of participants' artworks.

2.3. Sample size

The personnel involved in the assessment of this experiment are from the Faculty of Humanities and Arts. The physicians responsible for the diagnostic evaluation are from the Maternal and Child Health Hospital, which has a substantial number of ADHD-diagnosed children undergoing clinical treatment. The study sample comprised 50 children aged 7–10 years, all diagnosed with ADHD, with a final analytic sample of 41 (Table 2). The sample size estimation for the present study was informed by prior pilot data reported in Zhang et al.,³³ which demonstrated a medium-to-large treatment effect (Cohen's $d \approx 0.6$) of AI-assisted co-creative Drawing therapy on attentional symptoms in children with ADHD, as measured by the SNAP-IV-26 scale. Based on this effect size, we employed G Power 3.1 to conduct a priori sample size calculations for a two-tailed independent samples *t*-test, assuming $\alpha = 0.05$ and desired power $(1 - \beta) = 0.80$. The result indicated a required total sample size of approximately 45 participants (with balanced allocation). Participants were consecutively recruited from the outpatient clinic of Zhuhai Maternal and Child Health Hospital. Randomisation to the AI-assisted or traditional drawing arm used a computer-generated 1:1 sequence with allocation concealment, recruitment and randomisation are distinct from population sampling. Participants were consecutively recruited from the outpatient clinic. randomisation ensured equal

allocation between arms, but the sample was not population-probability-based. A post-hoc power analysis was subsequently conducted to assess the actual power under the same effect size assumption. The results indicated an achieved statistical power of approximately 0.73, which, while slightly below the conventional 0.80 threshold, remains acceptable for early-phase exploratory research.³⁴

2.4. Inclusion criteria

The sample consisted of 41 children aged 7–10 years with a confirmed diagnosis of ADHD (based on DSM-5 criteria) from the outpatient clinic of Zhuhai Maternal and Child Health Hospital (Table 2). These participants were divided into an intervention group (N = 19) and a control group (N = 22). Written consent was obtained from all guardians, and verbal assent was given by the participants themselves.

Inclusion criteria;;
5) Ability to speak in sentences consisting of at least one word;TIQ refers to Total Intelligence Quotient, measured using the Wechsler Intelligence Scale for Children (WISC-IV). A TIQ above 70 ensured participants had sufficient cognitive ability to complete the drawing tasks and follow instructions.

2.5. Interventions

Based on previous experimental designs in Drawing therapy,³⁵ the six themes for this study were decided upon following evaluations by a professional art professor, a psychology professor, and a physician (Fig. 1). Both groups conducted the sessions with professional assistance and adhered to the Hooked Behavior Model,³⁶ with each session centred around a specific theme (Table 1). Initially, the House-Tree-Person test was used as an introductory theme,³⁷ allowing participants to understand themselves and become familiar with the materials. The second

Table 1
Thematic design of the AI drawing intervention experiment.

Stage	Topics	Aspects of impact
Stage one	Scrawl	Obtain the child's base state. The child begins to become familiar with the material
Stage Two	Portraits	Demonstrate self-esteem and express personal preferences
	Family	Tap into your relationship and feelings with your loved ones
Stage Three	Feeling	Express feelings by showing your favourite animals
	Emotion	Inspire the imagination and express emotions
	Scenario	Improve detail capture and cognitive skills

phase focused on intimate relationships and emotions,³⁸ guiding participants to express their feelings. The third phase involved past experiences, encouraging participants to concentrate their attention and attempt to recall and recreate details. This phased thematic Drawing therapy has proven effective for children with ADHD.³⁹

The materials used were oil pastels suitable for school-aged children. These materials are widely used in Drawing therapy due to their ease of handling and excellent color-blending properties.⁴⁰ Oil pastels can also express a rich variety of styles through different levels of pressure applied during drawing.⁴¹ The intervention group utilized AI generation technology to process children's drawings. The artwork of all children in the intervention group was processed using Midjourney Vision 5, and all results were analyzed according to protocol. Psychotherapists encoded these drawings into keywords to establish a "visual diary." At the end of each session, the therapists and technicians entered keywords (Three to five) dictated by the children and their artwork into the AI software, which immediately generated new artwork that was then fed back to the children in the intervention group as a reward. During the enhancement phase, therapists reviewed the artworks with the children and assessed behaviors and emotions, guiding them into the next segment of therapy

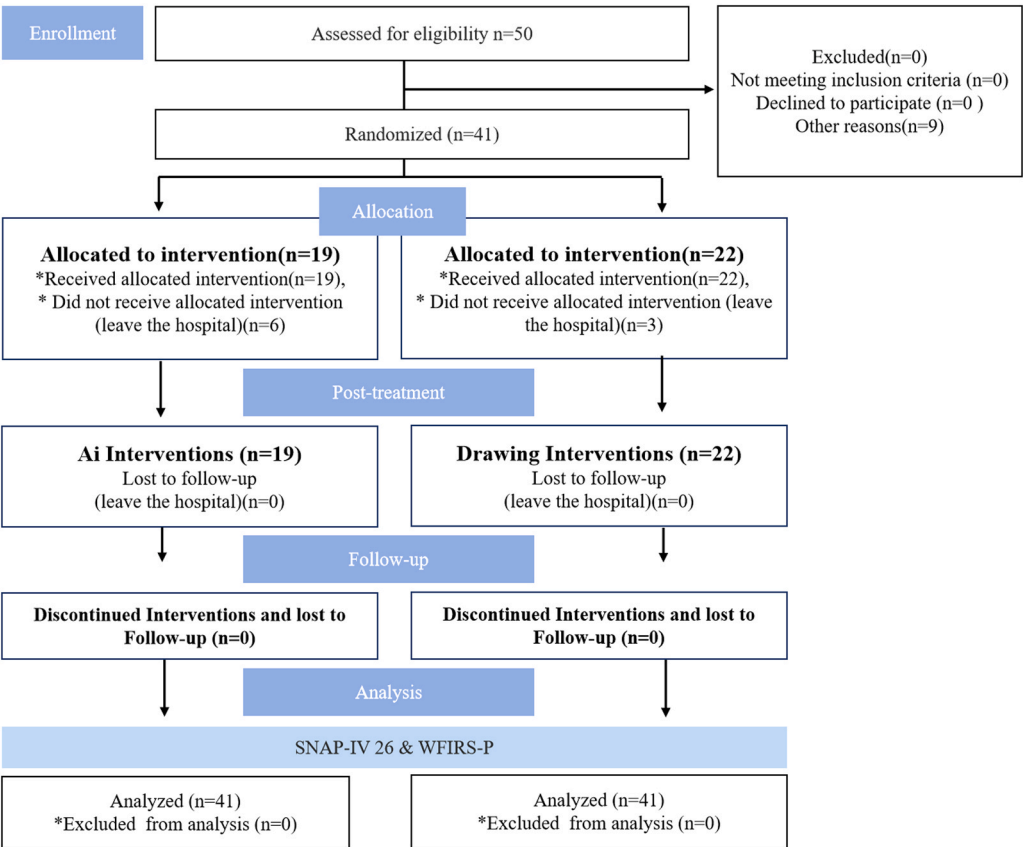


Fig. 1. CONSORT flow diagram of participant allocation.

tasks. And this group of children is only motivated to participate in the treatment by the rewarding feedback after drawing, and were not involved in using the interactive components of the AI at this stage. The control group focused on traditional drawing interventions, with the art therapist instructing weekly drawing sessions and providing verbal feedback. In other words, an AI-assisted drawing therapy mechanism of “child’s drawing-keyword coding-AI generation-feedback” was developed, whereby the child provided the material, and the therapist assisted in coding.

2.6. Outcome measure

The SNAP-IV-26 (Swanson, Nolan and Pelham-IV rating scales) was employed in this study as a tool for ADHD screening, auxiliary diagnosis, and assessment based on professional diagnostic principles.⁴² The evaluation encompasses three indicators: attention, hyperactivity/impulsivity, and oppositional defiant behavior. The Cronbach’s alpha coefficients for the scales were 0.903, 0.891, and 0.941, which are all greater than 0.80, and the reliability of the scale used in this case was good. The SNAP-IV-26 comprises 26 items rated on a 5-point Likert scale (0–4), where higher scores indicate greater symptom severity (total 0–104). Furthermore, the questionnaire’s average scores were utilized to determine whether there was an improvement in an individual’s ADHD symptoms. The mean scores and significance analysis post-intervention were utilized to detect improvements in ADHD symptoms, serving as the primary outcome measure.⁴³ The Weiss Functional Impairment Rating Scales were used to assess the impact of symptoms and behavioural or emotional issues on functional domains.⁴⁴ The scale includes six indicators: family, school, life skills, self-management, social activities, and risky activities. The scale includes 50 items across 6 domains, each rated on a 5-point Likert scale (0 = never impaired, 4 = severely impaired). Higher scores indicate greater functional impairment. The Cronbach’s alpha coefficients for the scales were 0.876, 0.886, 0.858, 0.812, 0.855 and 0.727; the reliabilities of the scales used in this case were good. Higher scores indicate greater impairment.⁴⁵ Total scores and significance analysis post-intervention were used to detect improvements in functional impairments associated with ADHD as a secondary outcome measure. We report summed raw scores (domain and total). The primary endpoint for inferential tests was the change-from-baseline (Δ), and post-intervention means are presented for descriptive context.

2.7. Data analysis

All analyses were conducted in IBM SPSS Statistics, version 29.0.2.0. Continuous variables are summarized as mean $\pm$ standard deviation (SD) and categorical variables as counts and percentages. The primary outcome measure was SNAP-IV-26. For SNAP-IV-26 and WFIRS-P, normality of change-from-baseline (Δ) and total scores within each group was assessed by Shapiro–Wilk tests and Q–Q plots, and homogeneity of variances was examined using Levene’s test. Between-group comparisons were performed with independent-samples *t*-tests on Δ scores (Welch’s *t*-test when variances were unequal); results are presented as mean differences with 95 % confidence intervals (CI). Within-group pre–post changes were examined with paired *t*-tests. All tests were two-sided with $\alpha = 0.05$.

2.8. Randomisation and masking

A total of 50 subjects were randomly assigned (1:1, 25 to the AI group and 25 to the control group). Nine subjects (6 to the AI group and 3 to the control group) dropped out before the intervention due to discharge/scheduling reasons, resulting in a total of 41 subjects (19 to the AI group and 22 to the control group) included in the analysis. Allocation concealment was ensured by an independent researcher. A blinded physician assessor collected SNAP-IV-26 and WFIRS-P at T1 and T2. Art

therapists delivered the interventions and, due to procedural differences, could not be blinded to group assignment; they did not conduct assessments nor access assessment data. Some parents/guardians became aware of group assignment during the study because of visible procedural differences; this partial unblinding is acknowledged in the limitations.

3. Results

3.1. Recruitment

Initially, 50 participants were recruited and randomly assigned equally to the experimental and control groups. The first 50 patients completed randomization (25 patients in each group). Nine patients dropped out before intervention (AI: 6/25 = 24.0 %; Drawing: 3/25 = 12.0 %), and 41 patients were included in the analysis (AI = 19; Drawing = 22). Notably, once interventions began, both groups demonstrated perfect retention with no further participant loss during the follow-up period (Fig. 1). This pattern suggests that while initial commitment may have varied between intervention types, both treatments maintained high engagement among participants who began the protocol.

3.2. Baseline data

Table 2 presents the baseline characteristics of the subjects and the comparison between groups. The average age was 8.12 years, with the majority being male ($n = 34$, 82.92 %), consistent with the gender distribution characteristics of children with ADHD.⁴⁶ The gender distribution was similar between the two groups, and the baseline characteristics of participants who completed the experiment were similar to those who withdrew midway. At follow-up, both groups showed similar SNAP-IV-26 total means. Because we report total (summed) scores rather than per-item averages, we do not assign severity labels here.

3.3. Primary outcomes

As shown in Table 3, both the AI interventions and drawing group demonstrated significant reductions in SNAP-IV-26 total scores within-group comparisons (AI: decreased from 75.63 ± 13.23 – 66.21 ± 9.64 , $t = 2.509$, $p = 0.017$; drawing intervention: decreased from 82.14 ± 19.65 – 67.95 ± 8.5 , $t = 3.103$, $p = 0.004$). Furthermore, within-group comparisons of the AI intervention revealed significant changes in attention ($p = 0.039$) and oppositional defiant behavior ($p = 0.025$). In between-group comparisons of SNAP total scores, the drawing group showed greater overall improvement ($p = 0.041$). According to the results of the between-group analysis of attention, the artificial intelligence group had a significant advantage in attention ($\Delta = -1.09$; $P = 0.035$), while the painting group performed better in oppositional defiant behavior ($\Delta = +1.706$; $P = 0.010$) and SNAP-IV-26 total score ($\Delta = +4.777$; $P = 0.041$). In addition, there was no significant difference

Table 2
Structure of the subject sample in the intervention and control groups.

Option	Quantity	Percentage	Baseline SNAP-IV-26	Baseline WFIRS
AI Intervention	19	46.34 %	81.47(2.16)	85(1.70)
Male	16	84.21 %	/	/
Female	3	15.79 %	/	/
Drawing intervention	22	53.66 %	82.14(2.35)	89.18(1.78)
Male	18	81.28 %	/	/
Female	4	18.18 %	/	/
Total	41	100 %	/	/
Male	34	82.93 %	/	/
Female	7	17.07 %	/	/

Table 3
Mean (standard deviation), change from baseline scores, and between-group change scores for descriptive statistical analyses and the independent samples *t*-test results for the two groups for SNAP-IV-26.

AI intervention	Drawing intervention					Comparisons between groups		
	ADHD symptoms	Baseline Mean±SD	Follow-up Mean±SD	t	P	Mean Difference (95 % CI)	Mean Change	p-value
Attention		41.05 ± 7.65	36.05 ± 6.73	2.137	0.039*	0.255, 9.745		
Hyperactivity-Impulsivity		19.68 ± 5.50	17.05 ± 3.55	1.753	0.088	-0.414, 5.677		
Oppositional defiant behaviour		14.89 ± 2.77	13.11 ± 1.85	2.343	0.025*	0.240, 3.339		
SNAP-IV – 26 Total		75.63 ± 13.23	66.21 ± 9.64	2.509	0.017*	1.804, 17.038		

Note: **p < 0.05; *p < 0.01; **p < 0.001 (two-sided *t*-tests). Mean change = (AI Δ – Control Δ); because higher scores indicate worse status on both scales, negative values favour the AI group.

Table 4
Mean (standard deviation), change from baseline scores, and between-group change scores for descriptive statistical analyses and the independent samples *t*-test results for the two groups for WFIRS-P.

AI intervention	Drawing intervention					Comparisons between groups		
	ADHD symptoms	Baseline Mean±SD	Follow-up Mean±SD	t	P	Mean Difference (95 % CI)	Mean Change	p-value
FAMILY		18.68 ± 4.46	15.16 ± 3.18	2.805	0.008**	0.976, 6.076		
SCHOOL ACTIVITIES		17.11 ± 4.42	14.42 ± 3.00	2.189	0.035*	0.197, 5.172		
LIFE SKILLS		19.11 ± 4.77	15.79 ± 3.61	2.415	0.021*	0.531, 6.101		
SELF-CONCEPT		6.26 ± 1.94	5.63 ± 1.74	1.057	0.298	-0.580, 1.843		
SOCIAL ACTIVITIES		11.53 ± 3.64	10.42 ± 3.04	1.015	0.317	-1.103, 3.313		
RISKY ACTIVITIES		12.32 ± 1.49	12.21 ± 1.58	0.211	0.834	-0.907, 1.118		
WFIRS-P Total		85 ± 18.04	73.63 ± 12.47	2.259	0.030*	1.164, 21.573		

Note: **p < 0.05; *p < 0.01; **p < 0.001 (two-sided *t*-tests). Mean change = (AI Δ – Control Δ); because higher scores indicate worse status on both scales, negative values favour the AI group.

between the two groups in hyperactivity-impulsivity ($\Delta=+0.082$; $P=0.076$), but there was a clear downward trend in painting (from 21.32 ± 5.89 – 16.77 ± 3.17 , $t=3.182$, $p=.003$). In brief, both interventions reduced ADHD symptomatology. The drawing group achieved larger gains in the SNAP-IV-26 total score and oppositional defiant behavior, whereas the AI arm showed a relative advantage for attention. Although mean changes were generally smaller with AI (except for attention), follow-up variability was lower across measures, indicating more consistent responses. Overall, the pattern suggests domain specificity: the AI intervention is comparatively effective for attention, while the traditional drawing intervention confers broader and larger benefits across the symptom spectrum, including oppositional defiant behavior and the SNAP-IV total score. Post-hoc analysis indicated an observed effect size of Cohen's $d=0.57$ for attention symptoms, consistent with the assumed effect size ($d=0.6$) in the a priori power calculation based on pilot data.

3.4. Secondary outcomes

On the WFIRS-P, the AI group showed a within-group significant reduction in total functional impairment (85.0 – 73.63 ; $t=2.259$, $p=.030$), whereas the drawing group exhibited a nonsignificant decrease ($t=1.759$, $p=.079$). The between-group comparison favored AI for the total score ($\Delta=-0.009$, $p=.047$; negative values indicate greater improvement with AI). At the domain level, the AI group achieved significant improvements in Family ($t=2.805$, $p=.008$), School Activities ($t=2.189$, $p=.035$), and Life Skills ($t=2.415$, $p=.021$), while the drawing arm did not reach significance in these domains. Risky Activities decreased in both groups without a significant between-group difference ($\Delta=-1.417$, $p=.082$). For Self-Concept, within-group changes were not significant, but the between-group effect favored drawing ($\Delta=+0.308$, $p=.036$). No significant between-group effects were detected for Life Skills, Social Activities, or School Activities ($p=.968$, $.997$, and $.058$, respectively). Overall, these secondary outcomes suggest clearer advantages of the AI intervention for global functioning and family-related domains, with a relative advantage of drawing for self-concept. All the painting tools used in this study and the images processed by AI will not have a negative impact on children with ADHD.

4. Discussion

Both groups of subjects showed significant effects on the main symptoms. This study not only observed the therapeutic potential of Drawing therapy for children with ADHD but also highlighted the advantages of AI as an auxiliary treatment, particularly in sustaining long-term engagement. Currently, many behavioural and psychological interventions face challenges in maintaining therapeutic effects due to the difficulty ADHD children have in focusing on task details. Although female patients are more prone to symptoms of inattention, with the assistance of AI, they gradually became immersed and independently completed multiple tasks.⁴⁷ However, the effect of AI intervention on hyperactivity-impulsivity indicators was not significant, possibly due to the excitement induced by AI-generated models during encouragement and feedback stages, which initially affected attention maintenance, possible short-term elevation of arousal levels due to the novelty of generated images (e.g., male subjects are more aroused), with a superimposed effect on hyperactive traits. In contrast, the control group adapted more quickly to traditional intervention methods. On one hand, three significant indicators in the control group confirmed that traditional art therapy under semi-participatory forms effectively alleviated externalising behaviours in ADHD children.⁴⁸ On the other hand, moderate-frequency art interventions have broad clinical application value in alleviating major ADHD symptoms, consistent with research findings.⁴⁹ Additionally, this study hopes that creative activities brought by AI-generated images provide children with a sense of satisfaction and

achievement, effectively reducing psychological stress and improving sustained attention for subsequent tasks. This also offers more opportunities for interaction with therapists, partially compensating for the limitations of single online art therapy.

AI-assisted art therapy has a positive impact on alleviating functional deficits in children with ADHD, particularly in family, school, and life aspects. These three dimensions reflect children's interaction needs with different relational levels and their self-management abilities. Firstly, ADHD children inherently have a weaker inclination to complete challenging tasks or perform tasks independently.⁵⁰ The involvement of AI makes them more engaged and attracted to the treatment, and the reward mechanisms enhance intrinsic motivation through AI feedback. AI optimizes chaotic images created by subjects, reducing anxiety associated with executing drawing tasks,⁵¹ and making them more willing to engage in family interactions. In addition, this study found that AI intervention may promote the therapeutic effect through its unique mechanism. AI's immediate mechanism significantly promotes children's "Flow" experience in art therapy by providing clear goals and immediate results, enhancing concentration and intrinsic motivation.⁵² The unique aesthetics and visual language of AI-generated artworks—such as instant image conversion and iteration—provide children with novel expressive mediums and "affordances",⁵³ which not only reduce the pressure of creation but also positively influence their emotional regulation and cognitive exploration, thereby increasing engagement in art therapy. Furthermore, research indicates that children with ADHD tend to maintain a state of low self-esteem during recovery,⁵⁴ primarily due to unmet basic functional optimization needs. Existing studies have demonstrated that psychological interventions combined with Cognitive Remediation Therapy (CRT) are effective in addressing functional deficits in children with ADHD.⁵⁵ In this study's intervention group, significant improvements were observed in family, school, and daily life functions. It is plausible that, for children with ADHD, visual stimuli are more readily processed by working memory than verbal instructions, making them more likely to interpret feedback as positive messages, thereby enhancing motivation to engage with family. Additionally, the generated images can serve as mediators for family discussions, reducing interaction barriers caused by the child's difficulties with verbal expression. AI-generated images also function as a reward mechanism, boosting children's self-esteem upon task completion.¹⁹ Lastly, the significant performance of the control group in adventurous activities can be attributed to children with ADHD effectively venting negative emotions through drawing.⁵⁶ In future applications, AI-assisted modalities are likely to serve as companions alongside clinicians and therapists, fostering increased patient engagement in art therapy.

This exploratory study demonstrated that AI-generated technology enhances traditional art therapy forms. However, preliminary observation in this RCT experiment also revealed additional impacts when AI is used as a primary auxiliary form, male subjects were more likely to become excited by its initial appearance. Considering that ADHD requires continuous treatment participation due to its long-term impact on school-age children,⁵⁷ further optimization is needed for the intervention form and frequency of AI-generated images.

5. Limitations and Future Work

A particularly notable limitation concerns the integrity of blinding within the study. Although assessors (physicians) were successfully blinded to group assignments, maintaining complete blinding for participants (children) and their parents/guardians proved challenging. As the AI-assisted intervention inherently involved distinct technological interactions (e.g., AI image generation and feedback) that visibly differed from traditional drawing therapy, some parents/guardians inevitably became aware of their child's group assignment several weeks into the intervention.⁵⁸ This partial unblinding introduces a risk of performance or observer bias, as parental expectations or knowledge

of the treatment arm could potentially influence their reporting of symptoms, functional impairments, or even their child's engagement. Moreover, the withdrawal of nine participants may reduce the external validity of the study because the characteristics of the dropouts (time conflict related to family engagement) may differ from those of the completers, resulting in limited generalization of the results. While this limitation is inherent to interventions with observable differences, future studies should explore strategies to enhance participant/parent blinding where feasible, or to rigorously quantify and account for potential expectancy effects. Furthermore, the study's relatively short follow-up period limits our ability to assess the long-term impact and sustainability of both AI-assisted and traditional drawing interventions; future research should consider longer follow-up periods to evaluate sustained effects.

Nevertheless, despite these methodological considerations, this research highlights how AI technologies can transform ADHD therapy for children by creating engaging environments that encourage self-expression and active participation.²⁵ While AI serves primarily as a supportive tool within art therapy frameworks, our findings reveal meaningful improvements in children's interactions and functioning across school, family, and daily settings. The accessibility of this AI-assisted approach makes it suitable for implementation in diverse clinical settings, including hospitals, rehabilitation centers, and community programs. However, we must carefully calibrate the frequency and intensity of AI exposure, particularly during initial sessions, to prevent overwhelming children who may be sensitive to novel stimuli. It is important to remember that artificial intelligence complements rather than replaces human expertise. The most promising path forward lies in thoughtful collaboration between AI systems and healthcare professionals, creating therapeutic experiences that meet children where they are—in comfortable, engaging formats that honor their unique needs and encourage healing through creative expression.

Ethics approval

This study follows the Helsinki Declaration and involves a randomized controlled trial. Ethical approval was obtained from the Ethics Committee of Macau University of Science and Technology (MUST-20230925001) and Zhuhai Maternal and Child Health Hospital (ZHFY-KXX-2023038, 2023 Ethics No. 175, 2023–11–08). And the trial is registered in the Chinese Clinical Trial Registry (ChiCTR2400089640).

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CRediT authorship contribution statement

Shumei Chen: Validation, Data curation. **Jue Wang:** Validation, Supervision, Resources, Conceptualization. **Jiajie Chen:** Validation, Methodology, Investigation. **Guanghui Huang:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Minglu Song:** Validation, Data curation. **Xuexing Luo:** Validation, Resources, Data curation, Conceptualization. **Qiang Chen:** Visualization, Validation, Supervision, Project administration, Methodology. **Xiang Zhou:** Validation, Resources, Project administration. **Dexin Li:** Supervision, Resources, Investigation, Formal analysis. **Yanrong Tang:** Validation, Investigation, Data curation. **Runqing Lin:** Writing – original draft, Visualization, Resources, Formal analysis, Data curation, Conceptualization. **Wenhua Yang:** Validation, Investigation, Data curation. **Aijia Zhang:** Writing – review & editing, Validation, Software, Methodology, Data curation. **Haotian Xu:** Validation, Investigation, Data curation.

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Declaration of Competing Interest

All the authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ctim.2025.103262.

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